



East West University

Assignment-3

Course Code:CSE 405

Course Title:Computer Networks

Semester:Fall 2025

Section:05

Course Instructor

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Submitted by

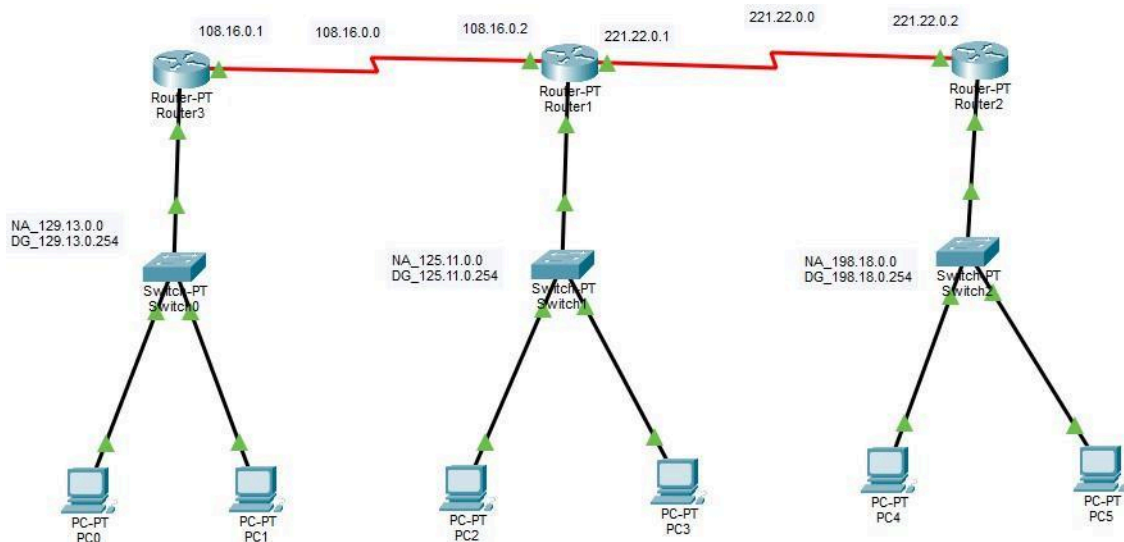
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Introduction:

This report outlines the design and implementation of routing configuration for the networks. The topology consists of three separate IP networks: Class A, Class B, and Class C. Each network connects to a dedicated router, which has switches that support end devices. To allow full communication among all networks, three dynamic routing protocols—OSPF, RIP, and EIGRP—are set up. All configurations are done using the Cisco IOS Command Line Interface (CLI). Successful ping results confirm correct inter-network connectivity.

Network Design:



The network has three routers arranged in a linear sequence. Router3, Router1, and Router2 communicate through two point-to-point networks. Router3 connects to Router1 via the 108.16.0.0/30 serial network, using 108.16.0.1 on Router3 and 108.16.0.2 on Router1. Router1 then connects to Router2 through the 221.22.0.0/30 serial network. Router1 uses 221.22.0.1 and Router2 uses 221.22.0.2. Each router also manages a

dedicated LAN with Class-C addressing. Router3 connects to the 129.13.0.0/24 LAN with a gateway of 129.13.0.254. Router1 connects to the 125.11.0.0/24 LAN with a gateway of 125.11.0.254. Router2 connects to the 198.18.0.0/24 LAN using 198.18.0.254 as the default gateway. Each LAN is expanded using a switch to two end devices, allowing PC-to-PC communication within the segment. By setting up static routes on each router to access the remote LANs through the correct next-hop addresses, all six PCs across the three networks can communicate smoothly through the serial backbone.

Router 3:

Rip:

```
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 129.13.0.0
Router(config-router)#network 108.16.0.0
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

En

Config

Router rip

Network 123.13.0.0

Network 108.16.0.0

Exit

exit

Eigrp:

```
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 1
Router(config-router)#network 129.13.0.0 0.0.255.255
Router(config-router)#network 108.16.0.0 0.255.255.255
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

En

Config

Router eigrp 1

Network 129.13.0.0 0.0.255.255

Network 108.16.0.0 0.255.255.255

Exit

exit

Ospf:

```
Router#
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 129.13.0.0 0.0.255.255 area 1
Router(config-router)#network 108.16.0.0 0.255.255.255 area 1
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

01:01:38: %OSPF-5-ADJCHG: Process 1, Nbr 221.22.0.1 on Serial2/0 from LOADING to FULL,
Loading Done
```

Config

Router ospf 1

Network 129.13.0.0 0.0.255.255 area 1

Network 108.16.0.0 0.255.255.255 area 1

Exit

exit

Router 1:

Rip:

```
Router>en
Router#comfig
Translating "comfig"...domain server (255.255.255.255)
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 125.11.0.0
Router(config-router)#network 221
% Unknown command or computer name, or unable to find computer address

Router(config-router)#network 221.22.0.0
Router(config-router)#network 108.16.0.0
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

```
Router con0 is now available
```

Eigrp:

```
Router(config-router)#network 125.11.0.0 0.255.255.255
Router(config-router)#network 108.16.0.0 0.255.255.255
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router eigrp 1
Router(config-router)#network 221.22.0.0 0.0.0.255
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 221.22.0.2 (Serial3/0) is up: new adjacency
```

Ospf:

```
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 2
Router(config-router)#network 125.11.0.0 0.255.255.255 area 1
Router(config-router)#network 108.16.0.0 0.255.255.255 area 1
Router(config-router)#network
01:26:55: %OSPF-5-ADJCHG: Process 2, Nbr 129.13.0.254 on Serial2/0 from LOADING to FULL,
Loading Done

% Incomplete command.
Router(config-router)#network 221.22.0.0 0.0.0.255 area 1
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

01:34:29: %OSPF-5-ADJCHG: Process 2, Nbr 221.22.0.2 on Serial3/0 from LOADING to FULL,
Loading Done
```

Router 2:

```
PT2005 processor: part number 0, mask 01
Bridging software.
X.25 software, Version 3.0.0.
4 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63489K bytes of ATA CompactFlash (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se3/0
Router(config-if)#ip address 221.22.0.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip address 198.18.0.254 255.255.255.0

% Invalid input detected at '^' marker.

Router(config-if)#ip address 198.18.0.254 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#exit
```

Rip:

```
Router>
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 221.22.0.0
Router(config-router)#network 198.18.0.0
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Eigrp:

```
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router eigrp
% Incomplete command.
Router(config)#network 221.22.0.0 0.0.0.255
^
% Invalid input detected at '^' marker.

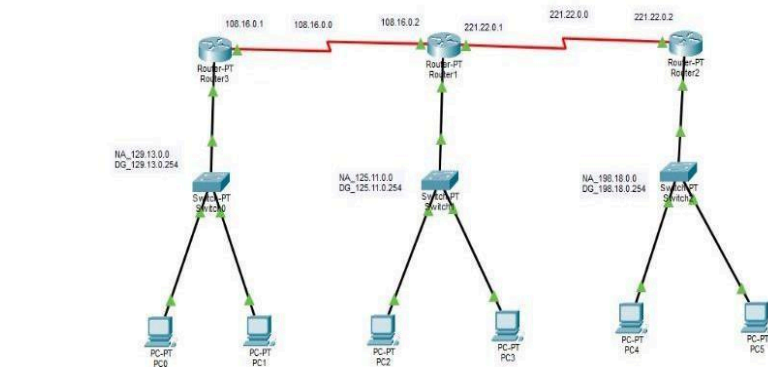
Router(config)#router eigrp 1
Router(config-router)#network 221.22.0.0 0.0.0.255
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 221.22.0.1 (Serial3/0) is up: new adjacency

Router(config-router)#network 221.22.0.0 0.0.0.255
Router(config-router)#network 198.18.0.0 0.0.0.255
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Ospf:

```
Router>
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 3
Router(config-router)#network 221.22.0.0 0.0.0.255 area 1
Router(config-router)#netwo
01:34:21: %OSPF-5-ADJCHG: Process 3, Nbr 221.22.0.1 on Serial3/0 from LOADING to FULL, Loading Done
% Incomplete command.
Router(config-router)#network 198.18.0.0 0.0.0.255
% Incomplete command.
Router(config-router)#network 198.18.0.0 0.0.0.255 area 1
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

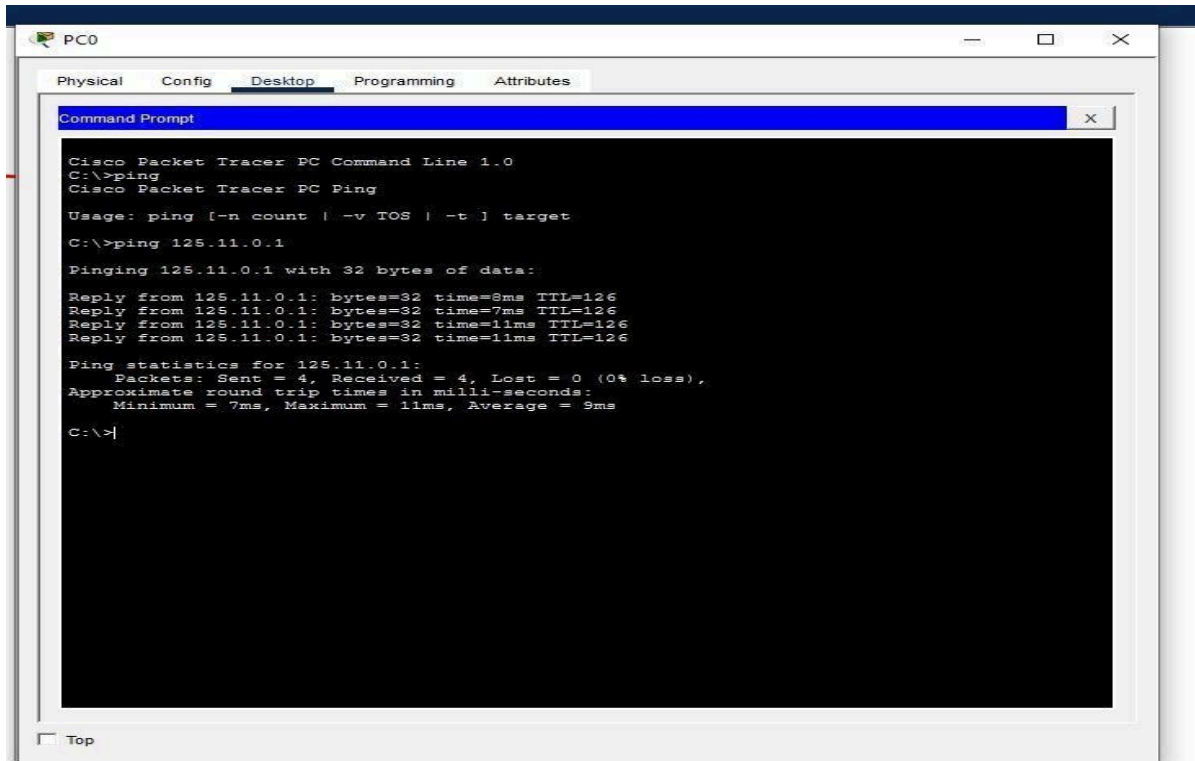
Packet Transfer:



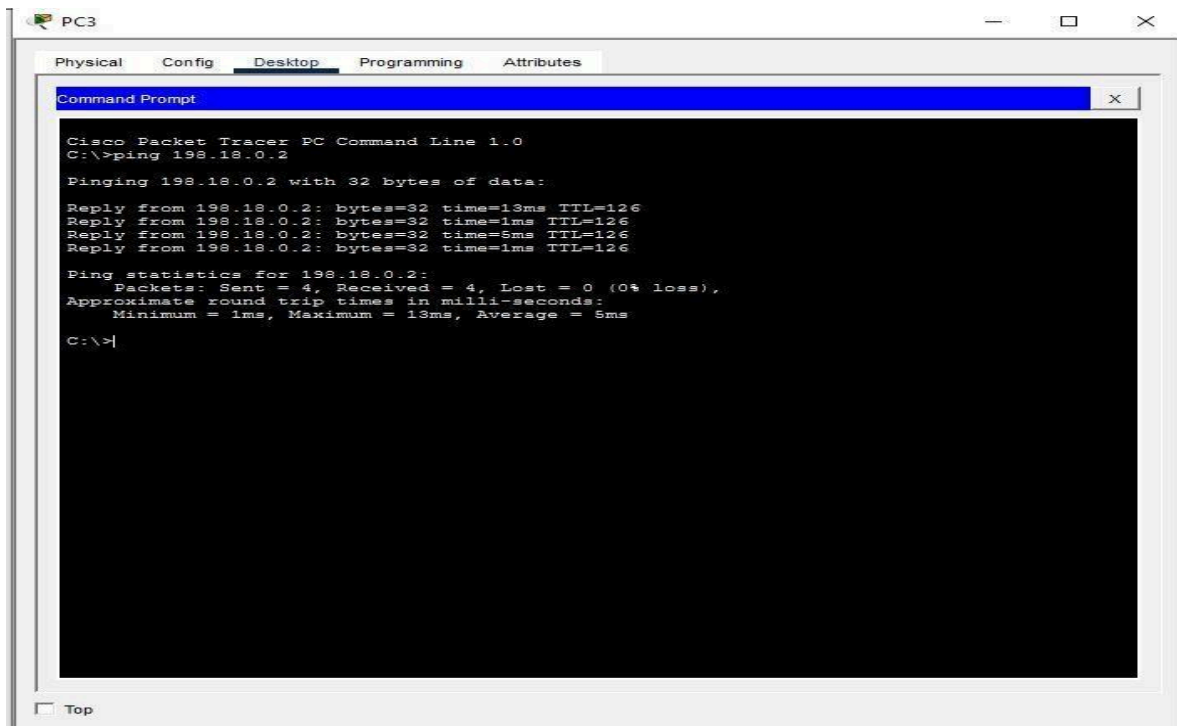
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Successful	PC1	PC2	ICMP		0.000	N	0	(e...)		(delete)
Successful	PC1	PC3	ICMP		0.000	N	1	(e...)		(delete)
Successful	PC1	PC4	ICMP		0.000	N	2	(e...)		(delete)
Successful	PC5	PC0	ICMP		0.000	N	3	(e...)		(delete)
Successful	PC4	PC3	ICMP		0.000	N	4	(e...)		(delete)

Packets travel from a PC to its default gateway router. The router checks its routing table and forwards the data through the connected routers to the right subnet. When the packet arrives at the destination network, the target PC gets it, and the reply takes the same route back.

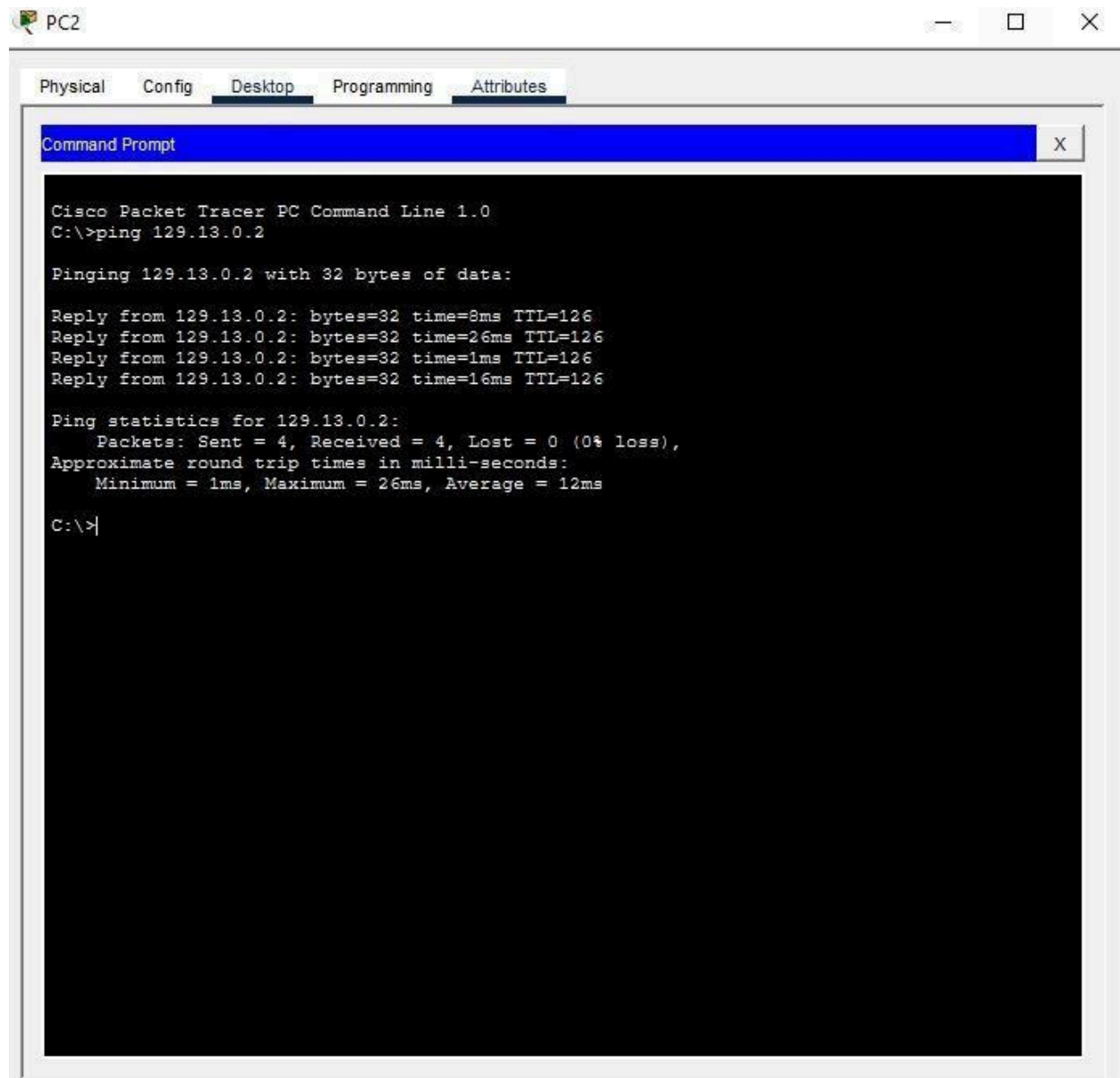
Ping: PC0 To PC2



PC3 To PC5



PC2 To PC1



Ping 129.13.0.2

Conclusion

In this project, we designed and set up a three-router network with connected LANs and serial links. We correctly configured the IP addresses, subnetting, and default gateways. We set up OSPF, RIP, and EIGRP, which allowed the routers to share routes and automatically build complete routing tables. We used verification commands and ping tests to confirm full end-to-end connectivity. The network was efficient, well-organized, and fully functional. It supported reliable and scalable communication across all segments.